

Guided Assignment 2

From spikes to computations

PN2 · Lecture 3 foundations · Lectures 4–5

Purpose. Build the intuition and written skills needed before attempting the existing PN2 practice test. This is ungraded, untimed practice, not an exam prediction. Scope follows PN2, with Lecture 3 encoding/decoding prerequisites and the temporal and transformation material in Lectures 4–5; the instructor’s exact Test 2 boundaries have not been verified.

How to work. Use separate paper. Predict first, calculate second, explain third. Notes and a basic calculator are allowed through Stage 7; no programming is needed. Keep exact answers where convenient. Start without the companion hints; consult one hint only after a serious attempt. Check the worked solution after completing a stage, then redo any missed part without looking. Say what changed in your understanding.

Suggested pacing. Three sittings: Stages 1–3, Stages 4–5, then Stages 6–8. Allow roughly 45–75 minutes per sitting, with more time if a prerequisite needs review. The estimates are flexible, not targets. Stage 8 is an independent readiness check; reserve the existing practice test until afterward.

Conventions. Neurons occupy rows of an activity matrix; training samples occupy columns. A rate is measured in Hz (s^{-1}). A spike has unit area. Time constants and time steps must use the same time unit. Angular frequency is $\omega = 2\pi f$ in rad/s. Fourier magnitude is $|X|$; power is $|X|^2$. A population’s radius describes its trained accuracy range, not an automatic clipping rule.

1. Reconnect encoding, activity, and decoding

Skill: Keep represented values, currents, activities, and decoder coefficients distinct.

For this small teaching example only, use rate neurons $a_i = \max(J_i, 0)$, with

$$J_1 = x + 1, \quad J_2 = -x + 1, \quad -1 \leq x \leq 1.$$

Activities here are in an arbitrary common rate unit, not the Hz values used later.

- 1a. Guided.** Complete the table at $x = -1, 0, 1$, listing J_1, J_2, a_1, a_2 . Sketch the two tuning curves. At $x = 0$, both neurons are active: why can the decoded value still be zero?

x	J_1	J_2	a_1	a_2
–1				
0				
1				

- 1b. Calculate.** Find coefficients satisfying $\hat{x} = d_1 a_1 + d_2 a_2 = x$ at all three samples. Use their sum and difference rather than a matrix inverse. Verify the result at $x = 1/2$. Is a negative decoder coefficient a negative firing rate?

- 1c. Generalize.** For n neurons, N samples and a d -dimensional variable, state the shapes of A, X, D in $\hat{X} = DA$. The noise-aware fitting objective is

$$\|X - DA\|_F^2 + \lambda \|D\|_F^2, \quad \lambda = N\sigma^2 > 0.$$

Complete $D(\text{_____}) = \text{_____}$, then solve for D symbolically. Explain why penalizing large coefficients can reduce sensitivity to activity noise while increasing clean reconstruction error.

Before moving on: You can explain the entire chain $x \rightarrow J \rightarrow a \rightarrow \hat{x}$ and check every matrix product's dimensions.

2. Turn a rate neuron into a spiking neuron

Skill: Connect current, voltage, spike timing, refractory time, and rate.

A normalized LIF neuron obeys $dv/dt = (J - v)/\tau_{RC}$ outside refractory periods. It spikes at $v = 1$, resets to zero, and holds there for τ_{ref} . Clamp negative voltages to zero. Use $\tau_{RC} = 20$ ms and $\tau_{ref} = 2$ ms.

- 2a. Predict.** Starting from $v(0) = 0$, sketch voltage for constant $J = 0.5, 1$, and 2 . Label the threshold, resets, and refractory intervals. Does approaching threshold imply reaching it in finite time?
- 2b. Guided derivation.** For constant $J > 1$, start from $v(t) = J(1 - e^{-t/\tau_{RC}})$. Set $v(t_{charge}) = 1$ and solve for t_{charge} . Add the refractory time to find the steady interspike interval and firing rate. Explain why the very first spike from rest has a different wait time.
- 2c. PN2 calibration.** Let $J = \alpha ex + J^{bias}$. Invert your rate expression to find the current $J(a)$ for a desired rate a . Use it to set a positive-encoder neuron to 40 Hz at $x = 0$ and 150 Hz at $x = 1$. Give expressions and approximate gain and bias. For its negative-encoder partner, predict both rates at $x = 0$, and which neuron is silent at $x = 1$. Sketch their relative spike density for $x(t) = \frac{1}{2} \sin(10\pi t)$; exact times are unnecessary.
- 2d. Simulation reasoning.** Write one forward-Euler voltage update for $\Delta t = 1$ ms, excluding reset/refractory handling. With $J = 2, v_k = 0.4$, calculate v_{k+1} . Give two reasons a one-second simulated spike count may differ from the steady-rate prediction. Which discrepancy can a smaller time step reduce, and which can remain?

Before moving on: You can sketch voltage and spikes separately, and explain why a 150 Hz rate does not guarantee exactly 150 recorded spikes in every one-second run.

3. Build intuition for time and frequency

Skill: Distinguish bandwidth, RMS, peak amplitude, and a real signal's Fourier coefficients.

Think of the notebook's periodic Fourier-coefficient construction over $T = 1$ s, sampled at $\Delta t = 1$ ms. Set DC to zero and use a cutoff of 8 Hz.

- 3a. Guided.** Find the number of samples, spacing between Fourier frequencies in Hz, and spacing in rad/s. If the coefficient at +3 Hz is $2 + 3i$, what must its -3 Hz partner be to produce a real signal? What coefficients are zero because of the mean and bandlimit constraints?
- 3b. Predict and sketch.** Sketch the ensemble-average magnitude versus ω , labelling both cutoff frequencies and units. A synthesized trace has RMS 0.8; write the rescaling that gives RMS 0.5. Does that guarantee every sample lies in $[-0.5, 0.5]$? Give a simple counterexample using a sine wave.
- 3c. Transfer.** Raise the cutoff from 8 to 16 Hz while keeping RMS 0.5. Sketch plausible short time traces before and after. Describe what changes in speed of variation and what is fixed about

amplitude. Must the exact peaks match? Explain why equal RMS does not imply equal difficulty for a synaptic filter to track the two inputs.

Before moving on: You can convert Hz to rad/s, impose conjugate symmetry, and distinguish RMS from a strict radius bound.

4. Decode spikes by adding synaptic responses

Skill: Understand convolution as superposition, then reason about smoothing and delay.

For a causal PSC use

$$h_n(t) = \begin{cases} t^n e^{-t/\tau} / c_n, & t \geq 0, \\ 0, & t < 0, \end{cases} \quad c_n = \int_0^\infty t^n e^{-t/\tau} dt.$$

- 4a. Guided.** Find c_0 and c_1 (integration by parts is useful). Optional: find c_2 . Sketch h_0 and h_1 at the same τ , labelling their values at zero and peak times. Explain what unit area preserves for a constant input; does it preserve every signal's RMS?
- 4b. Concrete superposition.** A neuron spikes at $t = 0$ and $t = \tau$. Write its filtered activity as two shifted kernels. For h_0 with $\tau = 10$ ms, calculate the activity at $t = 0^+$, $t = \tau^-$, and $t = \tau^+$. Use $e^{-1} \approx 0.368$. Include units and sketch the jumps and decays. These are filtered activity values, before multiplying by a decoder or synaptic weight.
- 4c. Units checkpoint.** Bin ℓ stores a spike count $b_{i\ell}$; the equivalent sampled impulse height is $a_{i\ell} = b_{i\ell} / \Delta t$. Kernel samples $h_k = h(k\Delta t)$ have units s^{-1} . Start from

$$s_{ik} \approx \Delta t \sum_{\ell=0}^k a_{i\ell} h_{k-\ell}.$$

Substitute the counts. Does the resulting count-based sum contain an extra Δt ? Explain why indiscriminately multiplying a rate decoder's output by Δt can give a scaling error.

- 4d. Predict before calculating.** With fixed decoders, compare increasing τ at $n = 0$ and increasing n at fixed τ . For each, explain residual spike ripple, rapid-signal distortion, and delay. If a fixed filter tracks an in-range 2 Hz input better than an in-range 10 Hz input, must the 2 Hz reconstruction be free of spiking noise?
- 4e. Small calculation.** For h_0 , use $|H(\omega)| = 1/\sqrt{1 + (\omega\tau)^2}$. At $\tau = 20$ ms, compute the gains at 2 Hz and 10 Hz. Relate these to your prediction. You may use $\pi \approx 3.14$.

Before moving on: You can turn spike times into a filtered trace and explain the tradeoff between reducing ripple and losing fast signal changes.

5. Understand what makes a filter optimal

Skill: Interpret an optimal filter as selecting stimulus-related response content.

An opponent pair has signed response $r = a_+ - a_-$. Let X and R be its input and response spectra. Angle brackets denote an ensemble average. Assume uncorrelated noise:

$$R = X + \eta, \quad \langle X\eta^* \rangle = 0, \quad S_X = \langle |X|^2 \rangle, \quad S_\eta = \langle |\eta|^2 \rangle.$$

5a. Guided. The optimal unconstrained linear filter is

$$H = \frac{\langle XR^* \rangle}{\langle |R|^2 \rangle}.$$

Expand numerator and denominator under the simplified response model to obtain a ratio in S_X, S_η . Explain why the response magnitude alone cannot tell you which frequencies are useful.

5b. Calculate and interpret. At one frequency $S_X = 9, S_\eta = 1$; at another $S_X = 1, S_\eta = 9$. Calculate both gains. For $S_X = 0$ and $S_\eta > 0$, what is the gain? If both powers vanish, why is direct division inappropriate, and what harmless filter choice could you make?

5c. Sketch. Suppose the input is bandlimited to 6 Hz and spiking noise extends outside that band. Sketch the qualitative support of input magnitude, response magnitude, and optimally decoded output magnitude. Distinguish a single random spectrum from an ensemble-average sketch. Does suppressing out-of-band noise imply exact reconstruction inside the band?

5d. Causality. Convolution can be written $\hat{x}(t) = \int h(u)r(t-u)du$. If $h(-5 \text{ ms}) \neq 0$, which response time contributes to the estimate at $t = 100 \text{ ms}$? Why can this be used on recorded data but not directly by a live synapse? Give a causal alternative and one cost.

Before moving on: You can explain both the spectral advantage and the possible biological limitation of an unconstrained optimal filter.

6. Transfer decoders and compute functions

Skill: Separate fitting, testing, and target functions; diagnose different errors.

Reuse Stage 1's rate population, with

$$A = \begin{bmatrix} 0 & 1 & 2 \\ 2 & 1 & 0 \end{bmatrix}, \quad X = \begin{bmatrix} -1 & 0 & 1 \end{bmatrix}.$$

6a. Guided target change. To decode $f(x) = 2x + 1$, form the target row F at these three inputs. Find $D^f = (p, q)$ that matches F exactly, and verify it at $x = -1/2$. State the general regularized function-decoder formula using $F \in \mathbb{R}^{q \times N}$. Here the output dimension q is unrelated to the coefficient name in the toy calculation.

6b. Representation limit. Try to decode x^2 with this same two-neuron rate population. Show why no fixed pair of coefficients matches all three samples. What is different between a linear decoder of nonlinear tuning curves and applying a linear map directly to x ? Explain why additional suitable nonlinear tuning curves can help.

6c. Rate-to-spike transfer. State how to use a rate-trained D with a spiking population and a common unit-area PSC: give a filtered activity equation and $\hat{x}(t)$. Does this promise exact moment-to-moment agreement with the rate prediction? State two conditions needed for a useful transfer.

6d. Diagnose. Match each observation to a likely mechanism and suggest a useful change: (i) a constant in-range input is consistently underestimated even after transients; (ii) a smooth input has an accurate average trend but visible jitter; (iii) a step rises gradually; (iv) error becomes large only when the target leaves the output population's trained range. Explain why increasing neuron count need not fix all four.

Before moving on: You can choose the training targets, reuse a decoder on new signals, and separate static distortion, jitter, filtering, and range error.

7. Turn decoded functions into network connections

Skill: Derive weights, choose output ranges, and distinguish addition from interaction.

- 7a. Guided derivation.** Population A has n neurons and a decoder $D^f \in \mathbb{R}^{q \times n}$. Population B has m neurons with encoder vectors $\mathbf{e}_i$, gains α_i and biases J_i^{bias} . Substitute $\hat{\mathbf{y}} = D^f \mathbf{a}$ into $J_i = \alpha_i \mathbf{e}_i^T \hat{\mathbf{y}} + J_i^{bias}$ and collect the coefficient of each a_j . Give w_{ij} , define the gain-scaled encoder matrix E , and state W and its shape. Where does the bias enter?
- 7b. Verify concretely.** With a scalar function decoder $D^f = (3/2, -1/2)$, gains $(2, 3)$, encoders $(+1, -1)$, and biases $(1, 2)$, calculate W . At presynaptic activities $(1, 1)^T$, compute postsynaptic currents both by decoding/re-encoding and by $W\mathbf{a} + \mathbf{J}^{bias}$. Identify the positive and negative weights.
- 7c. Range and time.** For $y = 2x + 1$, find the desired output ranges for (i) $x = t - 1$, $0 \leq t \leq 1$; (ii) $x = 0.2 \sin(6\pi t)$. Which stays within an output population trained on $[-1, 1]$? Suggest an adequate new radius for the other. Sketch an ideal step and a plausible decoded response, labelling filtering and jitter. Why should you fit over an input domain rather than refit for each particular waveform?
- 7d. Multiple inputs.** Draw a network for $z = 2y + 0.5x$ with separate scalar input populations. Label population dimensions and connection functions. Write the current of one receiving neuron as the sum of the two weighted activity contributions and its bias. How many times is the receiving neuron's bias added?
- 7e. Independent architecture.** Draw a network for $p = xy$ starting with separate scalar input populations. Include a population jointly representing (x, y) , label the coordinate-routing connections and the outgoing function decoder. Prove that a direct sum $g(x) + k(y)$ cannot equal xy throughout a two-dimensional domain, even if g and k are nonlinear. Can a nonlinear receiving neuron fix the problem if its only represented input is the single sum $x + y$? Use two pairs with the same sum and different products.

Before moving on: You can derive a connection rather than recall a weight formula, and decide when a joint representation is needed.

8. Independent readiness check

Skill: Retrieve, transfer, and explain with the scaffolding removed.

Close your notes and the solutions. No calculator is needed; this check is untimed. Use a new sheet. Give equations or sketches and a short explanation for each.

- 8a. A real, zero-mean input is bandlimited to 12 Hz. Sketch its ensemble-average Fourier magnitude against angular frequency. If its RMS is unchanged but its cutoff falls to 3 Hz, predict the time trace and the effect on tracking by a fixed causal PSC.
- 8b. Two LIF neurons have $J_{\pm} = 1 \pm x$, $\tau_{RC} = 20$ ms, $\tau_{ref} = 2$ ms. From rest, sketch both voltage traces and spikes at constant $x = -1/2$.
- 8c. Normalize $te^{-t/\tau}$ on $t \geq 0$, find the peak time, and explain why increasing τ can both help and hurt decoding.
- 8d. For $A \in \mathbb{R}^{30 \times 200}$ and $X \in \mathbb{R}^{2 \times 200}$, write the noise-aware identity decoder and its dimensions. With spike-count bins and continuous-kernel samples, write the filtered activity sum, with the correct time-step convention.
- 8e. A function decoder $D^f \in \mathbb{R}^{2 \times 30}$ feeds 50 neurons. Derive the weight matrix shape and its entries. Draw networks for $z = x - 2y$ and $p = xy$, labelling represented dimensions and connection functions.
- 8f. An output population is trained on $[-1, 1]$ but should represent $y = 3x + 1$ for $x \in [-1/2, 1/2]$. Find the required output range and distinguish this problem from synaptic lag.

Readiness rule. Mark each answer *independent*, *needed a hint*, or *not yet clear*. Any answer needing help sends you back to its earlier stage. You are ready to attempt the practice test when you can redo all six independently, with compatible dimensions, units, and causal explanations. This is a preparation criterion, not a guarantee about the real test.

Tutoring in this chat. Send one part with your reasoning and confidence. I can check the reasoning, give the smallest useful hint, and ask a follow-up before revealing a worked answer.